

**Before the
Federal Communications Commission
Washington, D.C. 20554**

In the Matter of

**SPACEX'S APPLICATION FOR ORBITAL
DATA CENTERS**

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File No. SAT-LOA-20260108-00016

March 5, 2026

To Whom It May Concern:

This comment on the SpaceX application for up to 1 million orbital data centres (ODCs)¹ is provided by Professor Hugh Lewis, University of Birmingham, United Kingdom.

Professor Lewis has researched topics related to orbital debris, space safety, and space sustainability for more than 25 years, is an internationally recognised expert in these topics, and is the author of several computational models for understanding and predicting the orbital debris environment. Through his career, he has used these computational models to provide quantitative evidence and technical advice to government, industry, and academia in support of orbital debris mitigation, safety, and sustainability objectives. Additionally, Professor Lewis is an expert in spacecraft systems engineering, having taught this topic at undergraduate, graduate, and professional (continuing development) levels for more than 20 years.

While the proposed ODC system has implications for on-ground casualty risks, the dark and quiet sky, atmospheric aerosols, clouds, and other environmental systems, this comment focuses on issues related to the orbital debris environment. It should be read in conjunction with comments submitted to address those other aspects.

1. The satellites would need to maintain a disposal failure rate below 0.0015% to meet the Commission's target of 0.5% for cumulative collision probability over a 15-year licence term.

In its order granting SpaceX's third modification application for Starlink, the Commission included some calculations to justify its reporting requirement on failed satellites². In those calculations, the Commission assumed a system of 4,408 satellites, a cumulative lifetime collision probability for a single satellite of 0.0001 and a disposal failure rate of 0.5%. It concluded that over a licence term of 15 years, 50 satellites might fail with a corresponding cumulative, or aggregate, collision probability of 0.5%, or 1-in-200. Assuming a satellite lifetime of 5 to 7 years, this corresponds to 3 disposal failures per year. Hence, the Commission required SpaceX to report if more than 3 disposal failures occurred in any year and, through this, implied that an acceptable or manageable cumulative collision probability across the Starlink NGSO system would be 0.5% for a 15-year period. Using this same process, it is possible to estimate the maximum disposal failure rate for a NGSO system of 1 million satellites or the maximum lifetime collision probability for a single satellite to achieve the cumulative collision probability of 0.5% over a 15-year licence term. Firstly, assuming the lifetime collision probability for a single satellite in the SpaceX ODC system is 0.0001, as per Starlink³, the maximum disposal failure rate would need to be below 0.0015% for the SpaceX ODC system to meet the cumulative collision probability threshold of 0.5% over a 15-year licence term, which is

¹ Space Exploration Holdings, LLC, Application for Authority to Launch and Operate the SpaceX Orbital Data Center System, ICFS File No. SAT-LOA-20260108-00016 (Call Sign S00798) (filed Jan. 30, 2026) (Application).

² SpaceX Third Modification Order, Adopted April 23, 2021, para. 63.

³ SpaceX July 7 Letter. SpaceX provided figures based on scenarios potentially applicable to its Starlink satellites. The 0.0001 value assumes a tumbling satellite at the higher end for the range of altitudes where Starlink satellites operate.

about 330 times smaller than the value assumed by the Commission for Starlink. It represents one disposal failure in about 66,000 satellites, or a post-mission disposal success rate of 99.9985%. From a spacecraft systems engineering perspective, this is already an improbable target to achieve. Given the required large size of the ODC satellites and the corresponding high likelihood of collisions with lethal non-trackable debris⁴, such a prescriptive target will be pushed further beyond reach. It is also probable that the single satellite lifetime collision probability is substantially underestimated for ODC satellites failing at altitudes above current Starlink altitudes, as residual orbital lifetimes can reach several thousand years at higher altitudes. Hence, the estimated 0.0015% failure rate is optimistic.

2. **Each satellite would need to remain below a lifetime cumulative collision probability of 1.5E-9 to meet the Commission's target of 0.5% for cumulative collision probability over a 15-year licence term.** Using the Commission's quantitative approach, as before, and assuming the disposal failure rate remains at 0.5%, the maximum lifetime collision probability for a single satellite would need to be 1.5E-9, or about 1-in-600 million, on average. SpaceX recently reported about 148,000 manoeuvres in six months for the Starlink NGSO system⁵ (as currently deployed), or an average of about 19 manoeuvres per Starlink satellite over the reporting period. This was in response to conjunctions each with a collision probability greater than 3E-7 (or about 1-in-3.3 million). In its review of the Starlink system, the Commission has already noted that Starlink satellites are able to manoeuvre, thereby reducing the collision probability to a level that is one-tenth of the manoeuvre threshold level (i.e., 3E-8)⁶. These observational data and submissions by SpaceX about the Starlink system indicate that just one mitigated conjunction event involving a SpaceX ODC satellite would exceed the required single satellite lifetime collision probability level of 1.5E-9. Hence, the required single satellite lifetime collision probability would be impossible to achieve even if the satellites were at the same altitude as Starlink. Given the much higher altitudes intended for the ODC satellites⁷, it is reasonable to assume an even higher lifetime cumulative collision probability per satellite, even at altitudes with relatively low spatial density.
3. **The deployment of 1 million satellites will lead to the accumulation of 10,000 failed satellites in low Earth orbits over a 15-year licence-term.** Assuming a launch system able to deliver 1 million satellites to orbit over a 5-year period (the assumed lifetime of a single satellite), the average deployment rate is 200,000 satellites per year. If 0.5% of these suffer a disposal failure, then 1,000 derelict, intact satellites will be added to the orbital environment annually. This neglects any satellites launched to replace failed satellites. The residual orbital lifetime of a failed ODC satellite in a circular orbit can be approximated as

$$t_L = \frac{H}{a\rho_a C_D V_a (A/m)} \quad (1)$$

where H is the atmospheric density scale height, a is the semi-major axis of the satellite, ρ_a is the atmospheric density, C_D is the satellite drag coefficient, V_a is the satellite orbital velocity (assumed to be on a circular orbit) and A/m is the satellite area-to-mass ratio. Assuming an area-to-mass ratio of 0.0735 m²/kg for a tumbling satellite⁸, failed ODC satellites will have a residual orbital lifetime of 1 year or less at altitudes below about 520 km under average solar activity conditions. Hence, altitudes below 520 km will not see any significant accumulation of failed satellites. However, above 520 km,

⁴ Boley, A. & Byers, M. Satellite mega-constellations create risks in low earth orbit, the atmosphere and on earth. Scientific Reports, **11**, article no. 10642. Available at: <https://www.nature.com/articles/s41598-021-89909-7>. (2021).

⁵ SpaceX Gen 1 and Gen 2 Semi-Annual Constellation Status Reports, December 31, 2025.

⁶ SpaceX March 16 Ex Parte (March 16, 2021).

⁷ Space Exploration Holdings, LLC, Application for Authority to Launch and Operate the SpaceX Orbital Data Center System, ICFS File No. SAT-LOA-20260108-00016 (Call Sign S00798) (filed Jan. 30, 2026) (Application).

⁸ Modification of Authorization for the SpaceX Gen2 NGSO Satellite System, Annex-C (SpaceX Gen2 Starship Satellite DAS input parameters), October 11, 2024.

the failed satellite population will grow at rates increasing with altitude. For example, at 800 km, the residual orbital lifetime would be about 100 years, meaning the population of failed satellites here might be expected to decrease by 1% per year. At 1,150 km, the residual orbital lifetime would be about 1,000 years and the failed satellite population would decrease by 0.1% per year. The average residual orbital lifetime of ODC satellites between 520 km and 2,000 km altitude is about 4,480 years. Assuming two-thirds of the ODC satellites are deployed to altitudes above 520 km, the population of failed satellites at these altitudes might therefore be expected to grow to 10,000 over a 15-year licence term, with a combined mass of 20 million kilogrammes assuming a single satellite mass of 2,000 kg. There would be substantial implications for safety, sustainability, and astronomy. If the failure rate is higher than 0.5%, there would be a corresponding increase in this population and its impacts.

4. **A disposal failure rate above 0.00025% will lead to a runaway fragment population.** The stability model by Lewis and Kessler (2025) provides an analytical framework for evaluating the proposed ODC system⁹. In this evaluation, only the population of failed ODC satellites needs to be considered. Although the population of failed satellites will increase at a constant rate, as shown above, the stability model assumes it is in equilibrium, i.e., unchanging. Hence, the stability model will provide an optimistic evaluation for the proposed ODC system. For this evaluation, 10,000 failed ODC satellites – roughly equivalent to 15 years of operation – are assumed to be distributed according to possible orbits for the SpaceX ODC system¹⁰ and maintained in equilibrium. The critical number of intact satellites needed for a runaway fragment population is

$${}_RN_i = \frac{4\pi a^3 \rho_a C_D V_a (A_f/m_f)}{N_0 W V \sigma_f}, \quad (2)$$

where N_0 is the number of fragments with sufficient mass to catastrophically breakup another intact satellite, (A_f/m_f) is the average area-to-mass ratio of those fragments, σ_f is the combined cross-sectional area between an intact satellite and a fragment, W is a parameter accounting for eccentric fragment orbits, and V is the average relative impact speed. Lewis and Kessler found that catastrophic, collisional breakups of intact objects typically produce about 112 fragments with sufficient mass to catastrophically breakup another intact satellite. A fragment that is about the size and mass of a 1U cubesat would have sufficient energy to catastrophically breakup a 2,000 kg intact satellite, such as the Starship version of Starlink. Assuming $N_0 = 112$, $W = 1.1$, $V = 7.5$ km/s⁹, $(A_f/m_f) = 0.017$ m²/kg for cubesat-sized fragments, $C_D = 2.2$, and $\sigma_f = 299.65$ m² using data submitted by SpaceX for the Starship version of Starlink (tumbling)¹¹, then the population of failed satellites will exceed the critical number – the runaway threshold – at all altitudes above 390 km. This condition is for the deployment of the ODC system into a pristine low Earth orbit environment (i.e., without the existing populations of satellites and debris) and represents the entire altitude distribution for the ODC system as per the FCC filing (Figure 1). The disposal failure rate would need to be about 2,000 times smaller than 0.5% for the failed satellite population to remain below the runaway threshold. This would be equivalent to a satellite post-mission disposal success rate of 99.99975%, which is likely beyond all current and future technological capabilities.

⁹ Lewis, H.G. and Kessler, D. Critical number of spacecraft in low earth orbit: a new assessment of the stability of the orbital debris environment. In Proc. 9th European Conference on Space Debris. Available at: <https://conference.sdo.esoc.esa.int/proceedings/sdc9/paper/305/SDC9-paper305.pdf>. (2025).

¹⁰ See one possible orbital distribution published by J. McDowell at <https://planet4589.org/space/con/conlist.htm>

¹¹ Modification of Authorization for the SpaceX Gen2 NGSO Satellite System, Annex-C (SpaceX Gen2 Starship Satellite DAS input parameters), October 11, 2024.

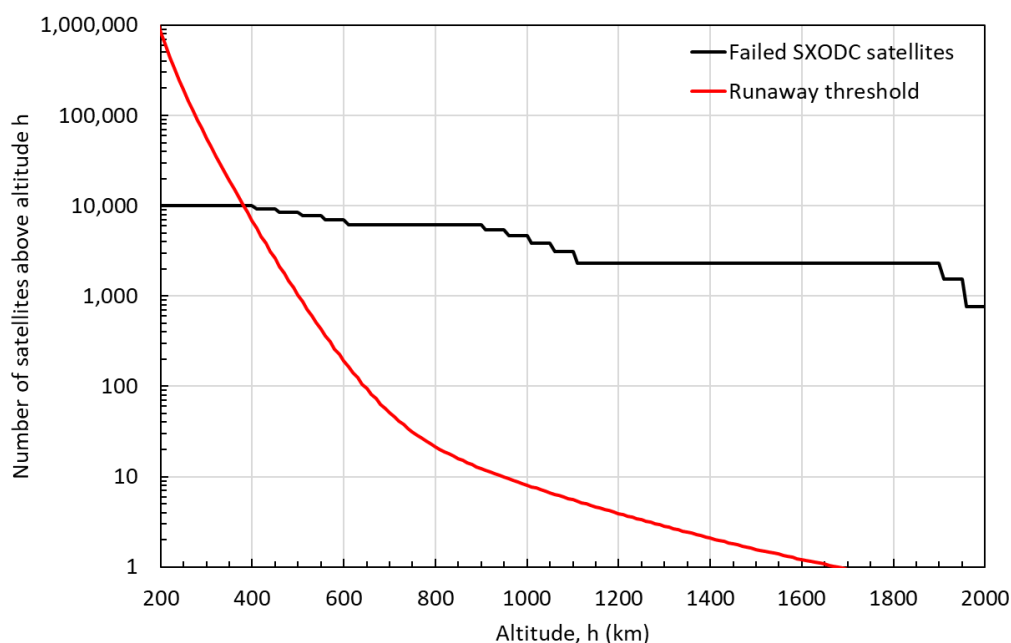


Figure 1. Regions where the proposed SpaceX ODC system would exceed the threshold for a runaway environment, assuming a disposal failure rate of 0.5%.

Even with an ODC system just 1% of the proposed size, a failure rate of 0.5% leads to a population of failed satellites that exceed the runaway threshold at altitudes above 690 km, corresponding still to a significant proportion of the orbital distribution for the ODC system (Figure 2). A failure rate about 20 times smaller than 0.5% would be needed for an ODC system that is 1% of the size of the one proposed, equivalent to a satellite post-mission disposal success rate of 99.975%.

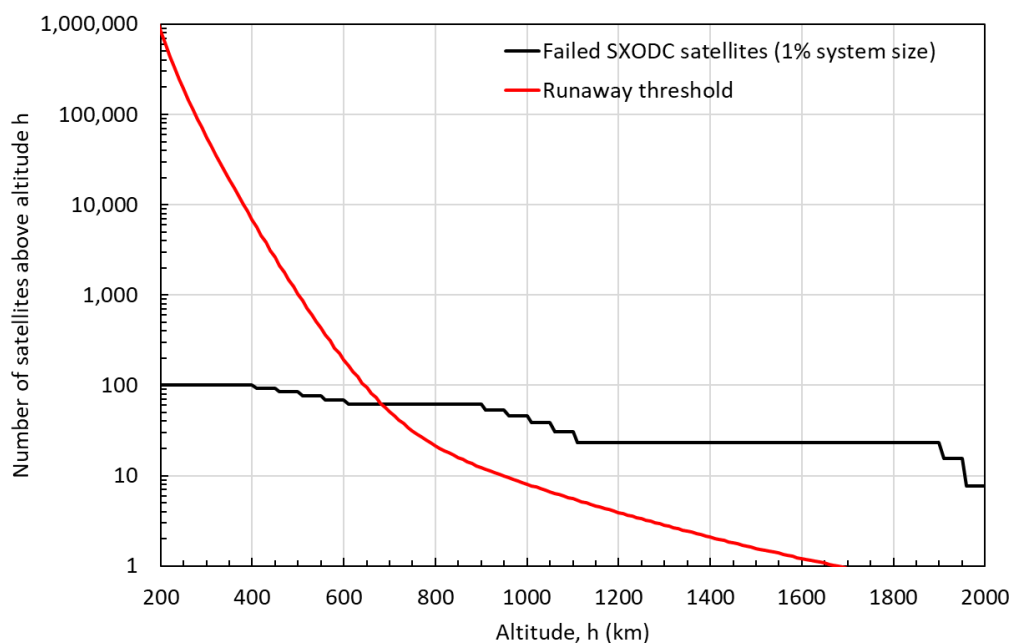


Figure 2. Regions where a system comprising 1% of the proposed SpaceX ODC satellites would exceed the threshold for a runaway environment, assuming a disposal failure rate of 0.5%.

Overall, the Commission's own quantitative approaches and the stability model by Lewis and Kessler indicate the proposal for a system of 1 million ODC satellites is contrary to the goals of safe operations and long-term sustainability of space activities. The technical requirements for achieving these goals are impossible to reach for such a system. Even the deployment of 1% of the proposed satellite number would

exceed the threshold for a runaway fragment population at altitudes above 690 km if the disposal failure rate remained at a realistic and achievable 0.5%. This outcome arises even without the consideration of the current satellite and debris populations in low Earth orbits, or plans for other large NGSO satellite systems. In other words, a system of 1 million ODC satellites – or even a system just a small fraction of the proposed size – would present irrevocable harms to the future use and exploration of space even if deployed into a pristine, debris-free orbital environment.

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